

Reasoning: Classification (Odd One Out)

Theory, 12 Solved Examples, and 25 Practice Questions with Answer Key

Part 1: Theory (Important Concepts of Classification)

1. What is Classification?

The word '**Classification**' means the process of sorting items, words, numbers, or letters into groups based on certain common properties or shared characteristics. In reasoning tests, the objective is to find the **'Odd One Out'** — the item that does not belong to the group formed by the rest.

2. Types of Classification Questions

- **Word Classification:** Based on meanings, categories, occupations, relationships, or scientific classifications (e.g., animals, fruits, professions).
- **Number Classification:** Based on mathematical properties such as prime numbers, squares, cubes, divisibility rules, or arithmetic operations (even/odd).
- **Alphabet / Letter Classification:** Based on alphabetical order, vowel-consonant relationships, or specific letter-skipping patterns.

3. Key Tips to Solve Classification Problems

- Examine all given options carefully and find a common thread or rule that applies to at least three of them.
- The option that violates this common rule or relationship is your correct answer.
- For numbers, always check prime numbers and squares/cubes first before looking at simple addition or subtraction.

Part 2: Solved Examples (12 Examples)

Example 1: Find the odd one out: (A) Apple (B) Mango (C) Potato (D) Banana

Solution: Apple, Mango, and Banana are fruits that grow on trees/plants above ground, whereas Potato is a root vegetable that grows underground.

Answer: (C) Potato

Example 2: Find the odd one out: (A) 27 (B) 64 (C) 125 (D) 144

Solution: 27 (3^3), 64 (4^3), and 125 (5^3) are perfect cubes. 144 is a perfect square (12^2), not a cube.

Answer: (D) 144

Example 3: Find the odd one out: (A) Carrot (B) Radish (C) Turnip (D) Tomato

Solution: Carrot, Radish, and Turnip grow underground (roots). Tomato grows above ground on a plant/vine.

Answer: (D) Tomato

Example 4: Find the odd one out: (A) 11 (B) 13 (C) 15 (D) 17

Solution: 11, 13, and 17 are prime numbers. 15 is a composite number divisible by 3 and 5.

Answer: (C) 15

Example 5: Find the odd one out: (A) Chair (B) Table (C) Sofa (D) Shirt

Solution: Chair, Table, and Sofa are pieces of furniture. Shirt is an article of clothing.

Answer: (D) Shirt

Example 6: Find the odd one out: (A) ACE (B) GIH (C) MOQ (D) SUW

Solution: In ACE (+2,+2), MOQ (+2,+2), and SUW (+2,+2), letters advance by +2. In GIH, H comes before I (G to I is +2, but I to H is -1).

Answer: (B) GIH

Example 7: Find the odd one out: (A) Copper (B) Iron (C) Aluminum (D) Brass

Solution: Copper, Iron, and Aluminum are pure chemical elements / metals. Brass is an alloy (mixture of copper and zinc).

Answer: (D) Brass

Example 8: Find the odd one out: (A) 36 (B) 49 (C) 81 (D) 63

Solution: 36 (6^2), 49 (7^2), and 81 (9^2) are perfect squares. 63 is not a square.

Answer: (D) 63

Example 9: Find the odd one out: (A) Delhi (B) Mumbai (C) Kolkata (D) Australia

Solution: Delhi, Mumbai, and Kolkata are major cities in India. Australia is an entire country/continent.

Answer: (D) Australia

Example 10: Find the odd one out: (A) Stream (B) River (C) Pool (D) Canal

Solution: Stream, River, and Canal contain flowing water. A pool contains stagnant/standing water.

Answer: (C) Pool

Example 11: Find the odd one out: (A) 12 - 48 (B) 14 - 56 (C) 15 - 60 (D) 16 - 68

Solution: In options A, B, and C, the second number is exactly 4 times the first ($12 \times 4 = 48$, $14 \times 4 = 56$, $15 \times 4 = 60$). For 16-68, $16 \times 4 = 64$, not 68.

Answer: (D) 16 - 68

Example 12: Find the odd one out: (A) Sparrow (B) Pigeon (C) Bat (D) Parrot

Solution: Sparrow, Pigeon, and Parrot are birds. Bat is a mammal that can fly.

Answer: (C) Bat

Part 3: Practice Questions (25 Questions)

Q1. (A) Tiger (B) Lion (C) Leopard (D) Cow

(A) Tiger

(B) Lion

(C) Leopard

(D) Cow

Q2. (A) 8 (B) 27 (C) 64 (D) 100

(A) 8

(B) 27

(C) 64

(D) 100

Q3. (A) Eye (B) Ear (C) Nose (D) Heart

(A) Eye

(B) Ear

(C) Nose

(D) Heart

Q4. (A) 17 (B) 19 (C) 21 (D) 23

(A) 17

(B) 19

(C) 21

(D) 23

Q5. (A) Rose (B) Jasmine (C) Lotus (D) Marigold

(A) Rose

(B) Jasmine

(C) Lotus

(D) Marigold

Q6. (A) DFH (B) MOQ (C) SUW (D) PRT

(A) DFH

(B) MOQ

(C) SUW

(D) PRT

Q7. (A) Mercury (B) Venus (C) Moon (D) Mars

(A) Mercury

(B) Venus

(C) Moon

(D) Mars

Q8. (A) 12 - 144 (B) 13 - 169 (C) 14 - 196 (D) 15 - 210

(A) 12 - 144

(B) 13 - 169

(C) 14 - 196

(D) 15 - 210

Q9. (A) Physics (B) Chemistry (C) Geography (D) Biology

(A) Physics

(B) Chemistry

(C) Geography

(D) Biology

Q10. (A) PEN (B) HEN (C) MEN (D) TEN

(A) PEN

(B) HEN

(C) MEN

(D) TEN

Q11. (A) 24 (B) 35 (C) 48 (D) 63

(A) 24

(B) 35

(C) 48

(D) 63

Q12. (A) Stomach (B) Brain (C) Liver (D) Blood

(A) Stomach

(B) Brain

(C) Liver

(D) Blood

Q13. (A) Hard : Soft (B) Long : High (C) Sweet : Sour (D) Cold : Hot

(A) Hard : Soft

(B) Long : High

(C) Sweet : Sour

(D) Cold : Hot

Q14. (A) AZ (B) BY (C) CX (D) DW

(A) AZ

(B) BY

(C) CX

(D) DW

Q15. (A) 125 (B) 216 (C) 343 (D) 513

(A) 125

(B) 216

(C) 343

(D) 513

Q16. (A) Inch (B) Foot (C) Yard (D) Quart

(A) Inch

(B) Foot

(C) Yard

(D) Quart

Q17. (A) 34 - 12 (B) 53 - 15 (C) 42 - 8 (D) 25 - 10

(A) 34 - 12

(B) 53 - 15

(C) 42 - 8

(D) 25 - 10

Q18. (A) Carrot (B) Radish (C) Ginger (D) Apple

(A) Carrot

(B) Radish

(C) Ginger

(D) Apple

Q19. (A) Flute (B) Guitar (C) Violin (D) Sitar

(A) Flute

(B) Guitar

(C) Violin

(D) Sitar

Q20. (A) 7 - 50 (B) 9 - 82 (C) 11 - 122 (D) 12 - 145

(A) 7 - 50

(B) 9 - 82

(C) 11 - 122

(D) 12 - 145

Q21. (A) Square (B) Circle (C) Rectangle (D) Triangle

(A) Square

(B) Circle

(C) Rectangle

(D) Triangle

Q22. (A) AEI (B) IOV (C) UAE (D) EIO

(A) AEI

(B) IOV

(C) UAE

(D) EIO

Q23. (A) Gold (B) Silver (C) Platinum (D) Diamond

(A) Gold

(B) Silver

(C) Platinum

(D) Diamond

Q24. (A) 45 - 9 (B) 36 - 6 (C) 28 - 7 (D) 64 - 8

(A) 45 - 9

(B) 36 - 6

(C) 28 - 7

(D) 64 - 8

Q25. (A) Father (B) Mother (C) Sister (D) Friend

(A) Father

(B) Mother

(C) Sister

(D) Friend

Answer Key for Practice Questions

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(D) Cow	Q8	(D) 15 - 210	Q15	(D) 513	Q22	(B) IOV
Q2	(D) 100	Q9	(C) Geography	Q16	(D) Quart	Q23	(D) Diamond
Q3	(D) Heart	Q10	(A) PEN	Q17	(C) 42 - 8	Q24	(A) 45 - 9
Q4	(C) 21	Q11	(B) 35	Q18	(D) Apple	Q25	(D) Friend
Q5	(C) Lotus	Q12	(D) Blood	Q19	(A) Flute	-	-
Q6	(A) DFH	Q13	(B) Long : High	Q20	(D) 12 - 145	-	-
Q7	(C) Moon	Q14	(AZ-DW...wait) (D)	Q21	(B) Circle	-	-